Shri Vile Parle Kelavani Mandal's
DWARKADAS J.SANGHVI COLLEGE OF ENGINEERING
(Empowered Autonomous College Affiliated to the University of Mumbai)

RETAIL SALES & PROFIT INTELLIGENCE STUDIO

Year 3, Sem V Computer Engineering

Business Intelligence and Analytics

By

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Guide:

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1. Introduction

Retail companies record thousands of sales transactions, but raw data does not show where the business makes or loses money. This project applies Business Intelligence and data-mining techniques to retail sales data to find profit drivers, loss-making products, customer groups and return patterns, and presents the results on an interactive dashboard.

Objectives:

- Clean and prepare the sales data for analysis.
- Measure business performance using KPIs.
- Predict profit, loss-making orders and future sales, and segment customers.
- Visualize the results and build a dashboard for decision-making.

Tools: Python, pandas, scikit-learn, Matplotlib, Seaborn, Plotly, Streamlit.

2. Data collection / Data description

The dataset used is the official **Sample - Superstore** dataset from Tableau Public (public.tableau.com/app/learn/sample-data). It contains the sales of a fictitious retail company in the United States and Canada from January 2023 to December 2026.

Sheet	Rows	Description
Orders	10,194	One row per order line (21 columns)
Returns	296	Order IDs that were returned
People	4	Regional manager of each region

Column(s)	Description
Order ID, Order Date, Ship Date, Ship Mode	Order and shipping details
Customer ID, Customer Name, Segment	Customer details (Consumer, Corporate, Home Office)
Country/Region, City, State/Province, Postal Code, Region	Location
Product ID, Category, Sub-Category, Product Name	Product details
Sales, Quantity, Discount, Profit	Sales value (USD), units, discount rate, profit/loss (USD)

The data has 5,111 orders, 804 customers, 1,862 products, 3 categories, 17 sub-categories and 4 regions.

3. Data pre-processing

- **Column names:** standardised (Country/Region to Country, State/Province to State).
- **Missing values:** none found.
- **Duplicates:** 2 duplicate rows removed (10,194 to 10,192 rows).
- **Data types:** dates converted to datetime; 438 US postal codes corrected by restoring the leading zero.
- **Validity checks:** no ship date before order date, no zero/negative sales or quantity, all discounts between 0 and 1. The 1,900 rows with negative profit were kept as valid losses.
- **Outliers:** 1,183 sales and 1,913 profit outliers flagged using the IQR method (not removed).
- **Data integration:** the Returns and People sheets were joined to the orders on Order ID and Region.
- **Feature engineering:** Year, Month, Quarter, Ship Days, Profit Margin %, Discount Band, Is Loss, Is Returned and Sales x Discount.

4. Mining models / analytics

4.1 Descriptive analytics

KPIs (total sales, profit, profit margin, orders, average order value and return rate) were calculated and analysed by category, region, segment, discount band and time.

4.2 Regression - predicting profit

Linear Regression and Random Forest were trained to predict the profit of an order line using Sales, Quantity, Discount, Sales x Discount, Category, Sub-Category, Region, Segment and Ship Mode (80/20 train-test split and 5-fold cross-validation).

4.3 Classification - predicting loss-making orders

Logistic Regression and Decision Tree (max depth 5) were trained to predict whether an order line makes a loss (Profit < 0), using the same features and a stratified 80/20 split.

4.4 Clustering - customer segmentation

Customers were grouped with K-Means (k = 4, chosen by the elbow method) on Recency, Frequency and Monetary (RFM) values.

4.5 Forecasting - monthly sales

A Linear Regression model with trend and month-of-year features was compared with a seasonal naive baseline. The last 12 months were used as the test set.

5. Results

Key performance indicators

Total Sales	Total Profit	Profit Margin	Orders	Avg Order Value	Return Rate
$2,326,154	$292,273	12.6%	5,111	$455	5.8%

Regression

Model	R2 (test)	R2 (5-fold CV)	MAE ($)
Linear Regression	0.80	0.62	35.45
Random Forest	0.67	0.59	19.54

Classification

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.94	0.94	0.72	0.82
Decision Tree	0.95	0.96	0.76	0.85

Clustering

Segment	Customers	Avg orders	Avg sales ($)
Champions	249	8.9	4,887
Loyal	202	6.8	2,683
Potential	262	5.0	2,021
Low Value	91	2.5	417

Forecasting

Model	MAE ($)	MAPE
Seasonal Naive	15,793	24.4%
Linear Regression	11,380	17.6%

Key findings

- Technology has the highest sales and profit (17.4% margin); Furniture has only a 2.6% margin.
- Tables, Bookcases and Supplies are loss-making sub-categories; 302 products make a loss.
- Discounts above 20% give negative margins (-15.3% for 21-40% and -77.4% for 41-80%).
- West is the most profitable region and Central the least.
- The West region has a 15.1% return rate, about three times the other regions (3.9% - 5.0%).
- Sales peak in November, December and September.

6. Visualization (Charts/plots)

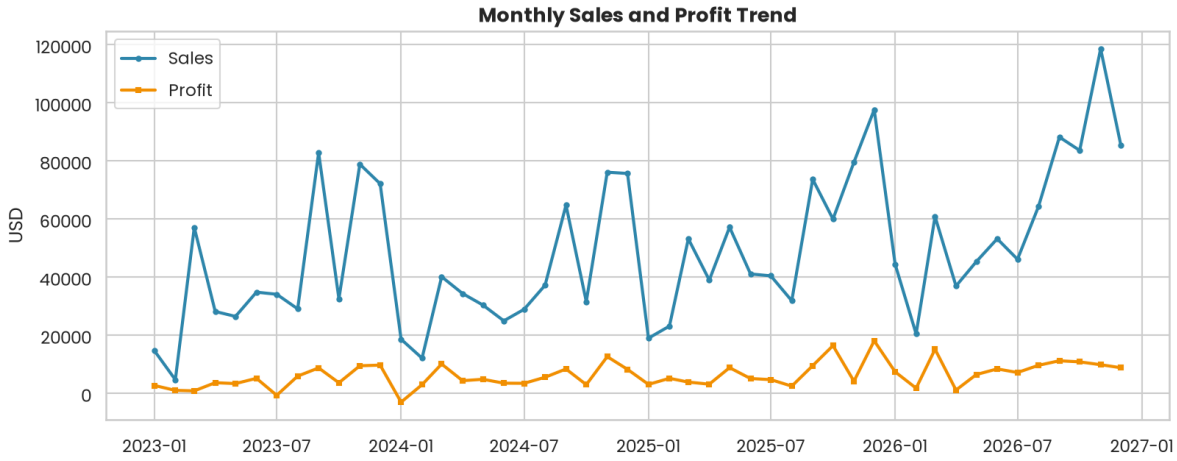


Figure 1: Monthly sales and profit trend

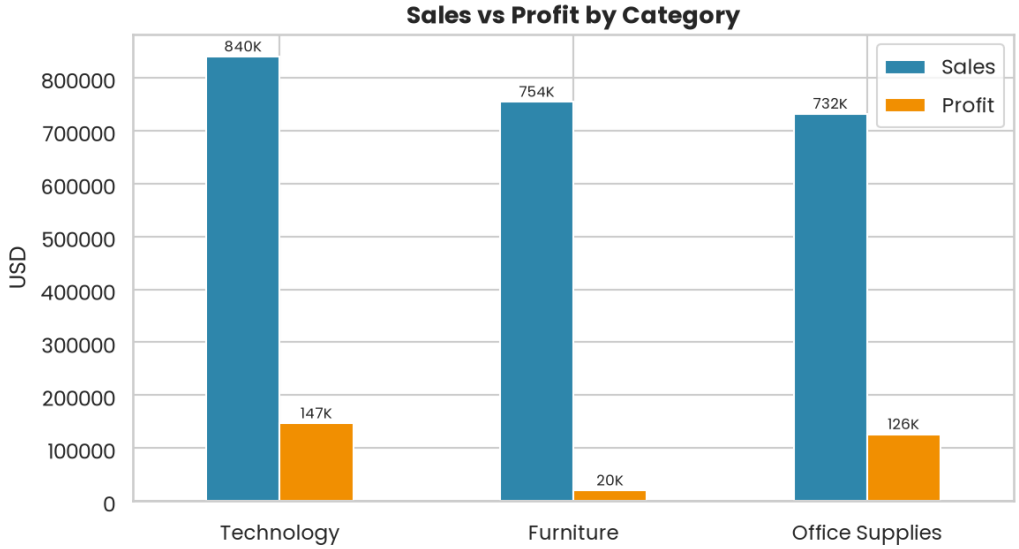


Figure 2: Sales vs profit by category

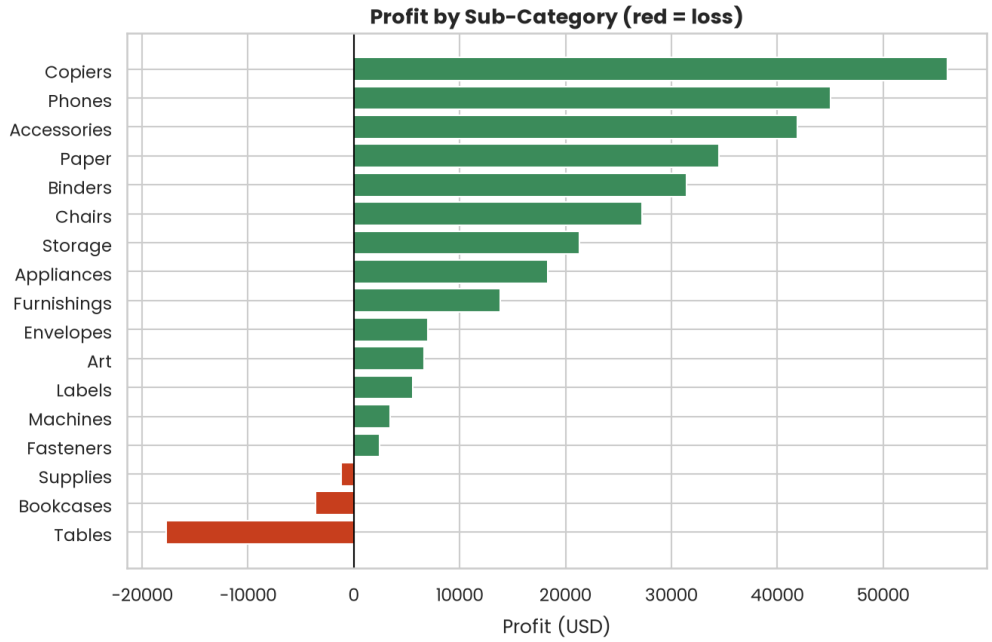


Figure 3: Profit by sub-category

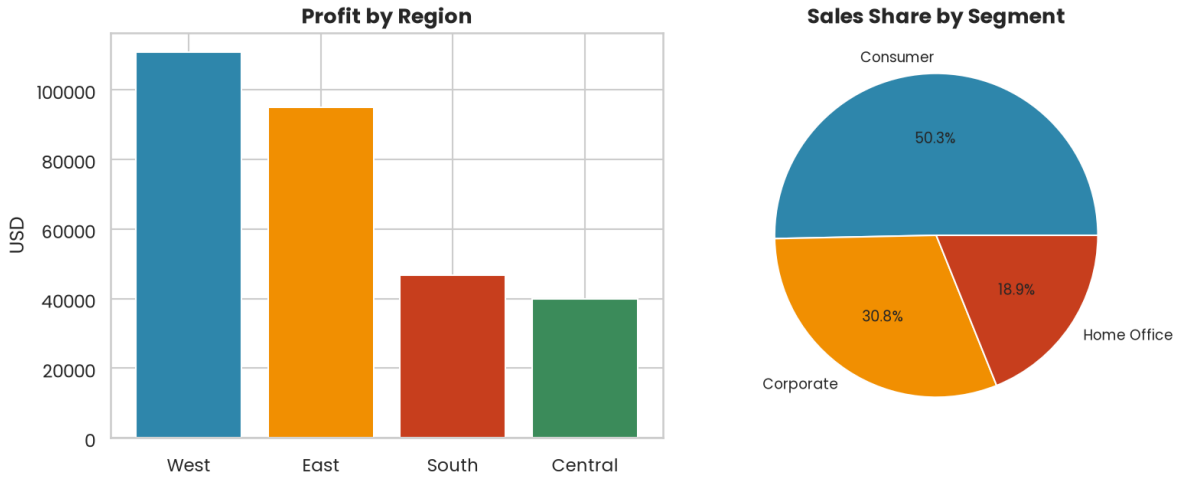


Figure 4: Profit by region and sales share by segment

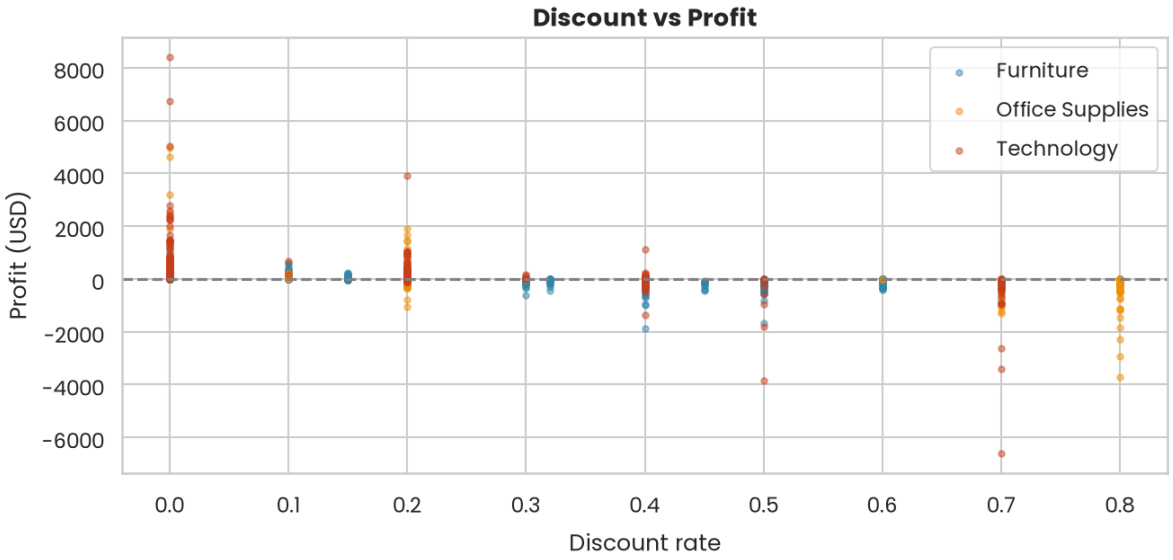


Figure 5: Discount vs profit

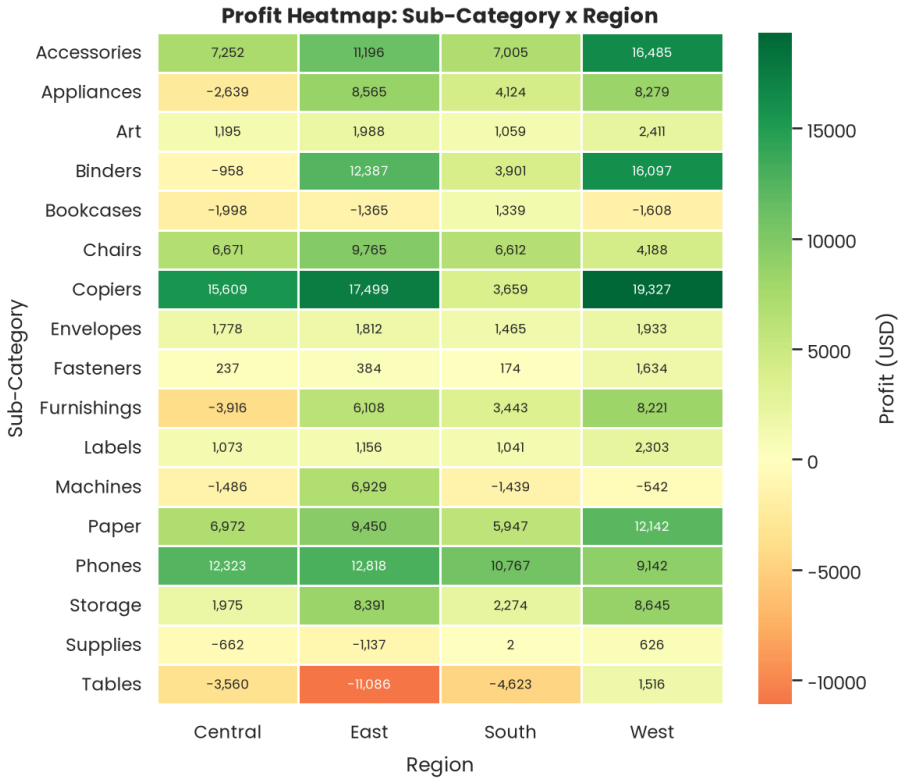


Figure 6: Profit heatmap by sub-category and region

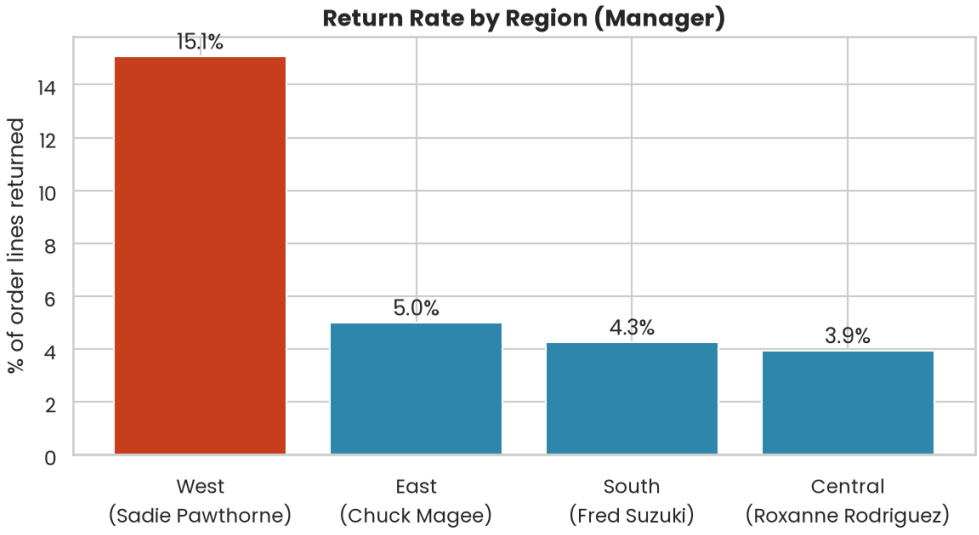


Figure 7: Return rate by region

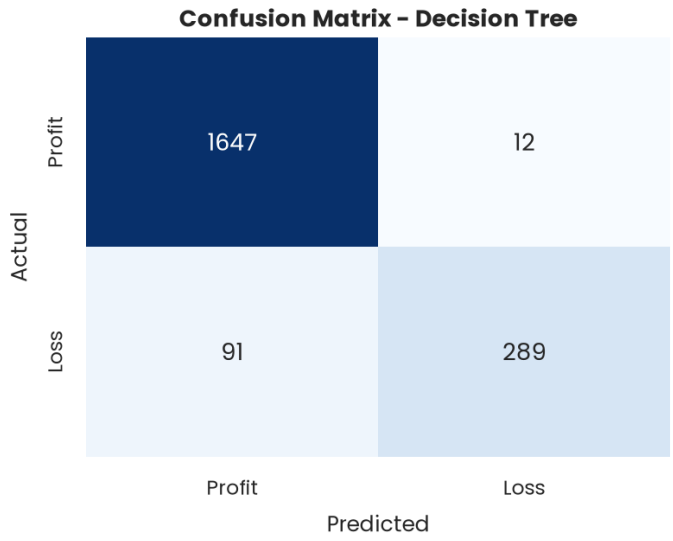


Figure 8: Confusion matrix of the Decision Tree

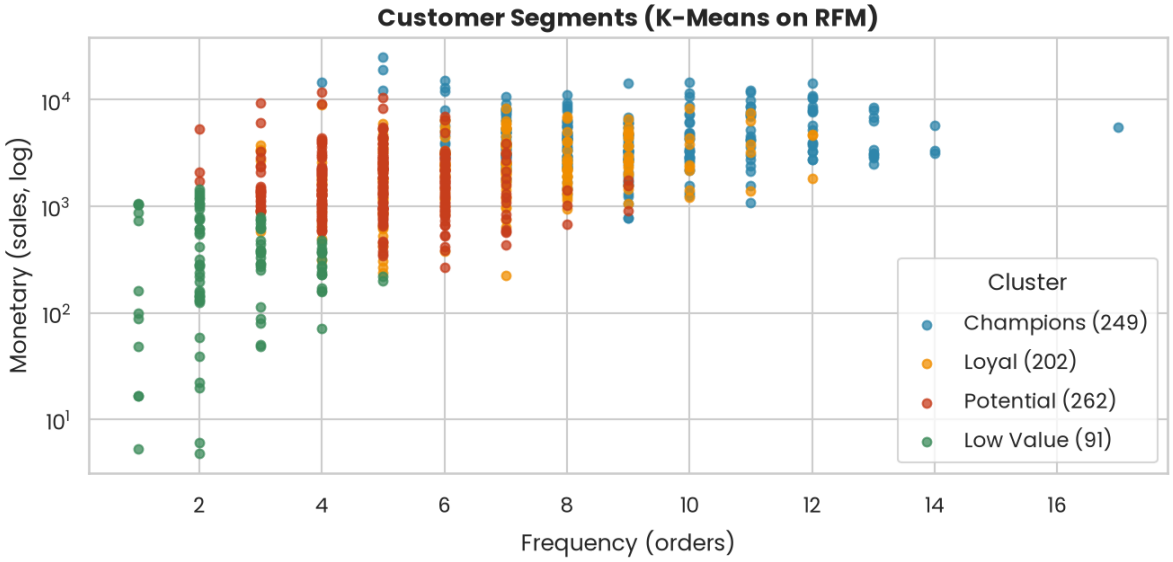


Figure 9: Customer segments (K-Means)

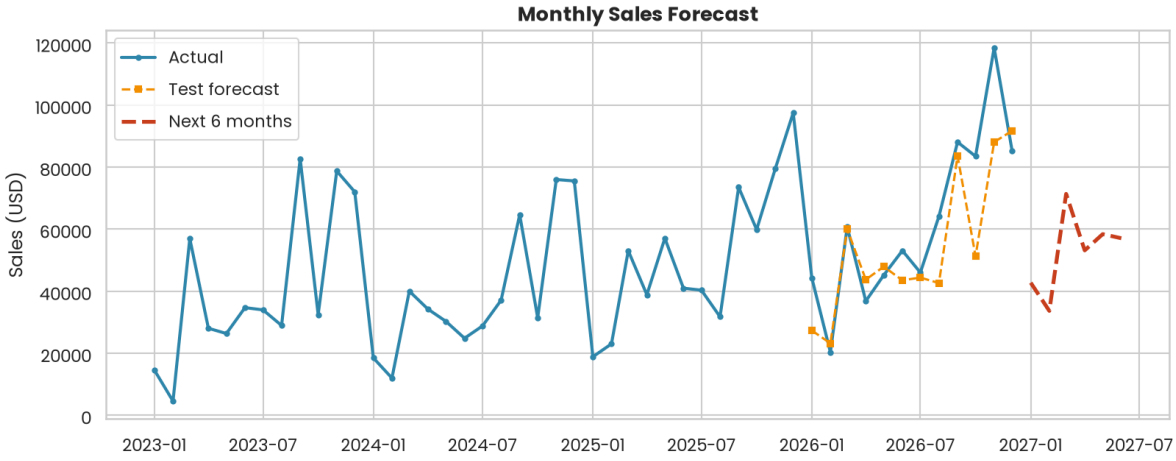


Figure 10: Monthly sales forecast

7. Conclusion

A complete Business Intelligence workflow (data collection, pre-processing, analytics, data mining and visualization) was applied to the Superstore retail data, with the results shown on an interactive dashboard.

- The Decision Tree predicted loss-making orders with 95% accuracy, and Linear Regression gave the best profit prediction (R2 = 0.80).
- High discounts are strongly linked to losses, so discounts above 20% should be limited.
- Loss-making products such as Tables and Bookcases need price or cost review.
- Champion and Loyal customers should be targeted for retention.
- The high return rate in the West region should be investigated.